

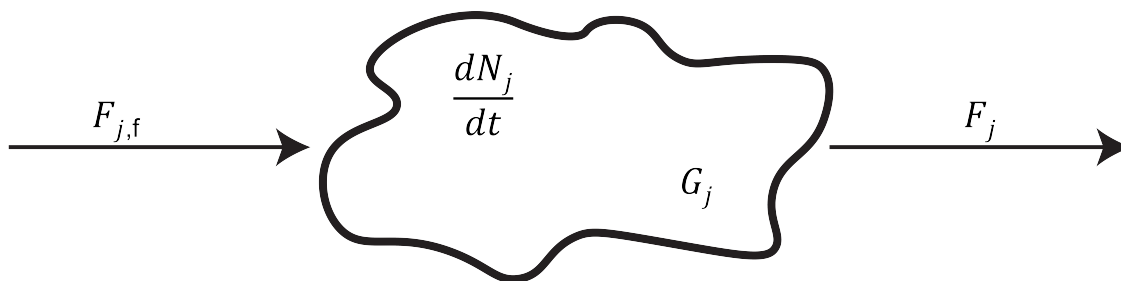
September 18, 2026

1 Material Balances II

What is Bond listening to right now?

This lecture continues discussion of material balances on reacting systems.

In [Notebook 13](#), we looked at a species balance on a reactor of arbitrary “size” and shape, where we considered that the “size” of the system might be described in terms of volume, mass of catalyst, interfacial area, or even number of catalytic “active sites” in the reactor. Our species balance on an arbitrary reactor:



Would give the following equation:

$$\frac{dN_j}{dt} = F_{j,f} - F_j + G_j$$

In other words, {accumulation} = {in} - {out} + {generation by reactions}. We finished [Notebook 13](#) with a brief presentation of G_j . We will pick this up here.

G_j is the **extensive** rate of generation (or consumption) of species j by chemical reaction. It represents how quickly species j is either consumed or produced by chemical reaction(s) that are occurring inside of our reactor in **extensive** units of **moles-per-time**. It is related to the rate of reaction, but it is not the same thing as the rate of reaction:

$$G_j \neq r_i$$

G_j is a **species specific term**. It is an **extensive** quantity that represents the amount of species j that is produced or consumed by chemical reaction in extensive units of moles-per-time. It will change depending on the size of the system and the particulars of the reactor you are using. In contrast, the rate of reaction, r_i , is a **reaction specific term**. It represents the rate of reaction i

that is occurring at the conditions inside of the reactor you are considering. Importantly, in Chemical Reaction Engineering, the rate of a chemical reaction, r_i , is always an *intensive* quantity that is scale and/or system independent. In terms of what controls the intensive rate of reaction, r_i :

$$r_i = f(T, P, \chi_j)$$

To expand that concept: once we decide on what reaction we're considering and what (if any) catalyst it uses, ***the rate of that reaction, r_i , is entirely determined by the temperature, pressure, and composition of the reacting mixture.*** The rate of reaction, r_i , does not depend on stoichiometry, nor does it depend on the type, size, or shape of a reactor. It might surprise you, but the rate of reaction also has no fundamental dependence on important aspects of reactor design, like mixing. The rate of reaction is always entirely determined by temperature, pressure, and composition. In contrast, stoichiometry, shape, size, type of reactor, extent of mixing, etc. will all influence the extensive rate of generation by chemical reaction, G_j :

$$G_j = f(T, P, \chi_j, \text{reactor type, reactor size, mixing, etc.})$$

These two quantities are related, but they are not the same thing. Next, we'll develop the relationship between the two.

1.1 A conceptual definition of a reaction rate

I have always liked the way that reaction rates are conceptually defined in Boudart's textbook *Kinetics of Chemical Processes*. They are not defined in terms of a species, rather, they are defined in terms of the amount of "reaction" that has occurred. Namely reaction rates are defined in terms of an extent of reaction, ε , which we know is an extensive quantity that describes how many "moles of reaction" have occurred. We want to transform that idea into a rate, specifically, into a rate of how many moles of reaction have occurred. At least from a dimensional standpoint, one could define an extensive rate of reaction i , $\bar{r}_i$ as:

$$\bar{r}_i = \frac{\Delta \varepsilon_i}{\Delta t} [=] \frac{\text{moles of extent of reaction } i}{\text{time}}$$

Since the extent of reaction i , ε_i , is an extensive variable, the above definition of $\bar{r}_i$ is an extensive rate of reaction, i.e., it varies with the scale of the system.

1.2 Intensive rates of reaction are more useful

Generally, we prefer to work with *intensive* reaction rates. The reason for this is that, fundamentally, a reaction rate should be a function of only (T, P, x_j) . It should not depend on the type of reactor we are using or the size of that system. This ensures that when I study kinetics in a small scale batch reactor, those results will apply to the same reaction happening in a large fluidized bed at ExxonMobil. Intensive reaction rates ensure the data are universal and that they will apply for a given (T, P, x_j) regardless of the reactor. This is a much more useful description of reaction rate than an extensive reaction rate that changes every time we change the reactor. Accordingly, in chemical reactor design, we will always work with intensive reaction rates that are normalized to a unit dimension of "size" or "scale" that describes our system. As we discussed in [Notebook 13](#),

what constitutes an appropriate dimensional scaling depends on the type of reaction. For homogeneous reactions that occur throughout the space occupied by the reaction mixture, volume is a good description of system size. We reach this conclusion because it is clear that increasing the volume of fluid within the reactor (for a fixed concentration of reactants) will increase the extensive quantity of reaction that is occurring. In other cases, catalyst mass, interfacial surface area, or even number of surface active sites may be appropriate bases for normalizing reaction rates. With this in mind, we present a few general conceptual definitions of intensive reaction rates using different types of normalizations that may be relevant in different types of reacting systems.

1.2.1 Homogeneous Reactions

For homogeneous reactions that occur throughout the space occupied by a fluid phase, volume is a good basis for normalizing reaction rates; hence, our convention for homogeneous systems is to report intensive reaction rates normalized to unit volume:

$$r_i = \frac{\Delta \varepsilon_i}{\Delta t \Delta V} [=] \frac{\text{moles of extent of reaction } i}{\text{volume} \cdot \text{time}}$$

We will use the notation convention presented in Scott Fogler’s *Elements of Chemical Reaction Engineering* wherein the rate of reaction i reported per unit volume will have the symbol r_i .

1.2.2 Heterogeneous Reactions

Catalytic reactions (e.g., in a packed bed) In cases where the reaction occurs on a catalyst, reactor volume is a much less relevant description of the reactor scale than the amount of catalyst present in the reactor. In practice, for catalytic reactions, we frequently will see intensive reaction rates reported per unit mass of catalyst. This makes sense because the *extensive* productivity of a catalytic reactor scales with the quantity (mass) of catalyst inside of the reactor much more so than it does the volume of space inside of the reactor. In this case, we define:

$$r'_i = \frac{\Delta \varepsilon_i}{\Delta t \Delta W} [=] \frac{\text{moles of extent of reaction } i}{\text{mass of catalyst} \cdot \text{time}}$$

Note that Scott Fogler’s textbook (*Elements of Chemical Reaction Engineering*) uses the symbol W for catalyst mass (catalyst “weight”). This has become standard notation for packed bed and fluidized bed reactors, so we’ll use it here. In our course, we’ll also use Fogler’s convention of using the symbol r'_i to indicate the mass-normalized rate of reaction i .

Reactions at interfaces between phases Sometimes, a reaction occurs only at an interface between two phases. In these cases, the extensive “scale” of the reaction is best described in terms of that interfacial area. When we have a reaction that occurs over a 2D surface or 2D interface between phases, it is often convenient for us to report reaction rates per unit area. Again, this is because the interfacial area is far more important than the total volume occupied in determining the overall, extensive performance of the reactor. In these cases, we might see a rate definition like this one:

$$r''_i = \frac{\Delta \varepsilon_i}{\Delta t \Delta A} [=] \frac{\text{moles of extent of reaction } i}{\text{interfacial area} \cdot \text{time}}$$

We will extend the prime notation used by Fogler as well as Dumesic (in *The Microkinetics of Heterogeneous Catalysis*) and use the symbol r_i'' to represent the rate of reaction i per unit area.

Reactions at specific active sites on a surface In [Notebook 13](#), we introduced the idea that catalysts are frequently comprised of expensive metals like Pt or Pd. In those cases, we almost never use a bulk sample of pure Pt or Pd. Instead, we disperse nanometer sized particles of Pt or Pd onto a high surface area carrier like silica or alumina. Frequently, the silica and alumina do nothing in terms of catalysis—they are there as a host for Pt or Pd nanoparticles. In these situations, the vast majority of catalyst mass or even catalyst surface area in the reactor is inert. Only a few exposed surface atoms of Pt or Pd are doing the catalysis, so the mass of catalyst and the surface area of the interface between the catalyst and the bulk fluid become poor descriptions of the size of the system. Instead, it is better for us to think of the reactor “size” in terms of the number of Pt or Pd atoms that are accessible in that system. This is a common convention in fundamental catalysis, where we usually report rates of reaction per active site:

$$r_i''' = \frac{\Delta \varepsilon_i}{\Delta t \Delta S} [=] \frac{\text{moles of extent of reaction } i}{\text{active site} \cdot \text{time}}$$

Usually, we quantify our active site in number of moles of active sites:

$$r_i''' = \frac{\Delta \varepsilon_i}{\Delta t \Delta S} [=] \frac{\text{moles of extent of reaction } i}{\text{moles of active site} \cdot \text{time}}$$

Which leads to a common (though not strictly rigorous) “cancellation” of moles to give a reaction rate in units of inverse time. This is called the turnover frequency of a catalytic reaction.

$$r_i''' = \frac{\Delta \varepsilon_i}{\Delta t \Delta S} [=] \frac{1}{\text{time}}$$

The notation here is consistent with that presented in Dumesic’s *The Microkinetics of Heterogeneous Catalysis*, where r_i''' represents the rate of reaction i per unit active site. It can be reduced to units of inverse time, and is also known as the **turnover frequency** of a catalytic reaction.

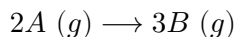
There are many types of catalysts, and they have many types of “active sites.” Above, we present the idea that a catalyst might be comprised of a metal dispersed on an inert carrier and that the active sites are accessible, surface metal atoms (not metal atoms buried in the bulk of a 3D particle). Some common examples of catalytic metals include Pt, Pd, Ir, Ru, Ni, Cu, and Os. Sometimes, we use metal oxides or mixed metal oxides as catalysts. In these cases, the active sites are often coordinatively unsaturated Lewis acidic metal cations (electron acceptors) like Al^{3+} ; Brønsted acidic protons, H^+ ; basic oxygen anions like O^{2-} ; or reducible metal cations like V^{5+} or Ti^{4+} . In all cases, the active site is likely a rare species on the surface of the catalyst.

1.3 Relating Production Rates to Reaction Rates

For the rest of this lecture, we’ll work with a homogeneous reaction and so consider the development of intensive reaction rates that are normalized per unit volume, i.e.,

$$r_i = \frac{\Delta \varepsilon_i}{\Delta t \Delta V} [=] \frac{\text{moles of extent of reaction } i}{\text{volume} \cdot \text{time}}$$

Consider a homogeneous reaction—let’s call it Reaction 1:



Under the conditions of our system, Reaction 1 is occurring with an *intensive* reaction rate of:

$$r_1 = 1 \frac{\text{moles of extent of reaction 1}}{\text{L} \cdot \text{min}}$$

Remember, we have defined reaction rates as properties of the reaction (not of a species), and they represent the rate of change in extent of reaction per unit time and size. In material balances, though, we are always interested in the rate that *species* are being generated by chemical reaction more than we are interested in the rate of the reaction itself. We can relate the intensive rate of reaction i , r_i , to the intensive production rate of species j , R_j , using stoichiometric coefficients. This is similar to what we do with extents of reaction in completing a mole table for equilibrium problems. For example, if we wanted to define the number of moles of A and B as functions of extent of reaction 1, ε_1 , we would write:

$$N_A = N_{A0} + \nu_A \varepsilon = N_{A0} - 2\varepsilon_1 \quad (1)$$

$$N_B = N_{B0} + \nu_B \varepsilon = N_{B0} + 3\varepsilon_1 \quad (2)$$

$$(3)$$

From this, you can see how stoichiometric coefficients allow us to relate a reaction specific property (like extent) to a species specific property (number of moles of A consumed). We can use an analogous approach to relate reaction-specific reaction rates, r_i , to species-specific production rates, R_j . For example, if we wanted to calculate the rate at which species A is produced, R_A , by reaction 1, we would consider that each time reaction 1 occurs, it consumes 2 moles of species A . R_A would therefore be defined as:

$$R_A = \nu_A r_1 = -2r_1 = -2 \frac{\text{moles of A}}{\text{L} \cdot \text{min}}$$

Similarly, the production rate of species B , R_B , is given by:

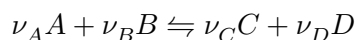
$$R_B = \nu_B r_1 = 3r_1 = 3 \frac{\text{moles of B}}{\text{L} \cdot \text{min}}$$

Notice that these quantities, R_j , are defined per unit size (volume in this case), so these capital R production rates, R_j are intensive quantities, just like reaction rates, r_i . Intensive production rates for homogeneous reactions will usually have the following units:

$$R_j [=] \frac{\text{moles of j}}{\text{volume} \cdot \text{time}}$$

1.3.1 Generalizing the Concept

Based on that example above, we can conclude that for any, generic single reaction:



That is occurring with an intensive reaction rate, r_i . We would define the intensive production rate of a species participating in that reaction as:

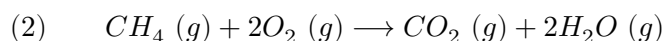
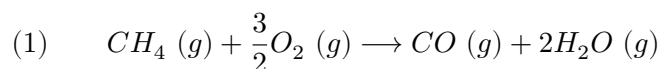
$$R_j = \nu_j r_i$$

Conversely, in some situations, we are able to estimate the rate of reaction (which we can't measure directly) from the production rate of a species (which we can measure directly):

$$r_i = \frac{R_j}{\nu_j}$$

1.3.2 Multiple Reactions

There is no restriction that a species is only participating in one reaction in a given reactor. In fact, the more realistic scenario is that each species is actually participating in multiple reactions. This means that each species is being consumed or produced by multiple chemical reactions, and so when we define the production rate, R_j , we actually need to account for the rate of consumption from many chemical reactions. We'll do this with an intuitive example and then present a general equation. Consider partial and complete combustion pathways that occur during methane oxidation:



These are different reactions, and their rates will have different dependencies on temperature, pressure, and composition. In other words:

$$r_1(T, P, x_j) \neq r_2(T, P, x_j)$$

Because of this, we have to account for the rate of each reaction separately when we quantify the production rate for each species. Conceptually, it's pretty straightforward. Analogous to our solution of multiple reaction equilibria using mole tables, we simply add up the consumption or production rate for each species in each reaction. So for methane (CH_4), we find:

$$R_{CH_4} = -1r_1 - 1r_2$$

And we can do the same thing for every species participating in these two reactions.

$$R_{O_2} = -\frac{3}{2}r_1 - 2r_2 \quad (4)$$

$$R_{CO} = +1r_1 + 0r_2 \quad (5)$$

$$R_{CO_2} = +0r_1 + 1r_2 \quad (6)$$

$$R_{H_2O} = +2r_1 + 2r_2 \quad (7)$$

$$(8)$$

From that exercise, it makes sense that we can formally calculate an intensive production rate for any species, j , by summing up the intensive production rate for that species in each reaction that is occurring in the system. That results in the following definition of intensive production rates:

$$R_j = \sum_{i=1}^{N_R} \nu_{i,j} r_i$$

Here, the summation goes over all reactions ($i = 1$ to N_R , where N_R is the number of reactions occurring in the system). The stoichiometric coefficient, $\nu_{i,j}$, is the coefficient of the j^{th} species in the i^{th} reaction, and r_i is the rate of the i^{th} reaction. You'll notice that these species production rates are still intensive and, for homogeneous systems, have the following units, just as they did for a single reaction:

$$R_j [=] \frac{\text{moles of } j}{\text{volume} \cdot \text{time}}$$

1.3.3 Connecting Intensive production rates, R_j to Extensive production rates, G_j

Recall that a species balance on a reactor is fundamentally an extensive balance that changes with the size of the system and has units of moles/time:

$$\frac{dN_j}{dt} = F_{j,f} - F_j + G_j$$

The **basic** idea of relating an intensive property of a system to an extensive property of a system is that we simply scale the intensive property to the size of the system.

A Simple Example Consider that we have a homogeneous reaction, $A \rightarrow B$, that is occurring in a 1 Liter reactor. We will again call this “Reaction 1”, and we will set the concentration of A at 1 mole per liter, and there is no B initially in the system. At these conditions, the intensive rate of Reaction 1 is:

$$r_1 = 1 \frac{\text{moles of extent of reaction 1}}{\text{L} \cdot \text{min}}$$

Based on our discussion above, the intensive production rates for A and B would be:

$$R_A = -1 \frac{\text{mol}}{\text{L} \cdot \text{min}} \quad (9)$$

$$R_B = 1 \frac{\text{mol}}{\text{L} \cdot \text{min}} \quad (10)$$

$$(11)$$

And to get the extensive production rates for A and B, we'd multiply those intensive rates by the size of the system, which is 1L here:

$$G_A = -1 \frac{\text{mol}}{\text{min}} \quad (12)$$

$$G_B = 1 \frac{\text{mol}}{\text{min}} \quad (13)$$

$$(14)$$

Scaling to 100L If we scale this system to a 100L reactor with identical temperature, pressure, and composition, we find that it has an identical intensive rate of reaction, which makes sense because reaction rate only depends on (T, P, x_j) , which is identical in the two reactors:

$$r_1 = 1 \frac{\text{moles of extent of reaction 1}}{\text{L} \cdot \text{min}}$$

Similarly, intensive production rates are identical in the 100L reactor and the 1L reactor:

$$R_A = -1 \frac{\text{mol}}{\text{L} \cdot \text{min}} \quad (15)$$

$$R_B = 1 \frac{\text{mol}}{\text{L} \cdot \text{min}} \quad (16)$$

$$(17)$$

It is only when we get to calculating extensive rates—which we do by multiplying intensive rates by the size of the system—that we see a difference:

$$G_A = -100 \frac{\text{mol}}{\text{min}} \quad (18)$$

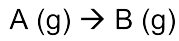
$$G_B = 100 \frac{\text{mol}}{\text{min}} \quad (19)$$

$$(20)$$

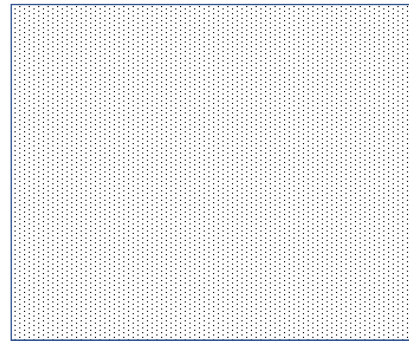
For a Homogeneous Reaction:

Reaction occurs throughout volume
Volume is a good description of “size”

Example:



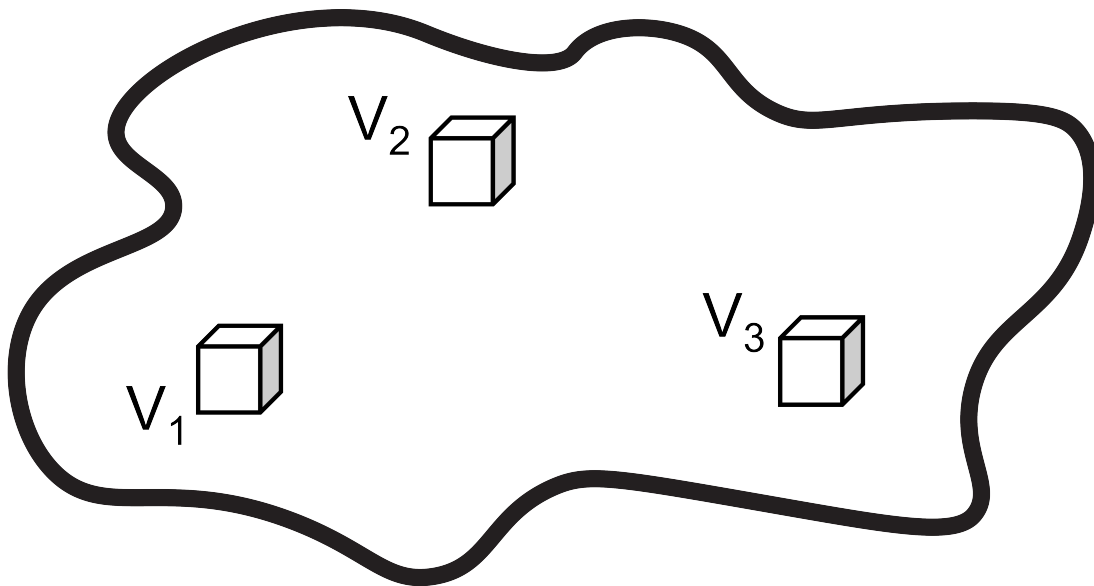
1L Reactor
1 Mole of A
 $C_A = 1 \text{ mole/L}$
 $r = 1 \text{ mol/L/min}$
 $RA = -1 \text{ mol of A/L/min}$
 $RB = 1 \text{ mol of B/L/min}$
 $GA = -1 \text{ mol of A/min}$
 $GB = 1 \text{ mol of B/min}$



100L Reactor
100 Moles of A
 $C_A = 1 \text{ mole/L}$
 $r = 1 \text{ mol/L/min}$
 $RA = -1 \text{ mol of A/L/min}$
 $RB = 1 \text{ mol of B/L/min}$
 $GA = -100 \text{ mol of A/min}$
 $GB = 100 \text{ mol of B/min}$

1.3.4 Doing this for a real reactor...

What we did above, directly multiplying an intensive production rate by the volume of the system is something we can only do if the intensive reaction rate is *the same* throughout the reactor volume. This is not always true. Consider our arbitrary reactor:



$$V = \sum_{k=1}^m V_k$$

This reactor has a total volume of V . We can further conceptually subdivide that volume into many small volumes, each with a volume of V_k . If we sum up all of those volumes, we would get the total volume of the system:

$$V = \sum_k V_k$$

Now we consider potential differences between the small volume element at position 1, V_1 , and the one at positions 2, V_2 , and 3, V_3 . We know that rates of reaction are a function of only temperature, pressure, and composition. So if we have a single reaction i occurring in this reactor, we could define the reaction rate inside of any volume element k , $r_{i,k}$ as:

$$r_{i,k} = f(T, P, x_j)_k$$

We have no way to guarantee that the temperature, pressure, and compositions are equal at all of the positions in our reactor, so it is entirely possible that:

$$(T, P, x_j)_1 \neq (T, P, x_j)_2 \neq (T, P, x_j)_3$$

Because $(T, P, x_j)_k$ fully determine the reaction rate within volume element k , it follows that it is possible for:

$$r_{i,1} \neq r_{i,2} \neq r_{i,3}$$

If this is true, then it also means that production rates may be spatially non-uniform throughout the reactor:

$$R_{j,1} \neq R_{j,2} \neq R_{j,3}$$

It is important to note that we are still only considering a single reaction—the subscripts 1, 2, and 3 refer to **positions** in the reactor. The implication is that, without making some specific assumptions about our reactor, we **cannot** guarantee that reaction rates and production rates are spatially uniform everywhere inside of the reactor. Since we can't guarantee spatially uniform rates, we **cannot** simply multiply the intensive production rate of a species by the reactor volume to get the extensive production rate. It's slightly more difficult...

We start with our individual volume elements in the system, V_k . We will assume that each of these elements is small. Moreso, we will assume that it is sufficiently small that we can say the temperature, pressure, and composition **inside** of a single element, V_k are spatially uniform throughout that small volume element. If this is true, then we can say that the intensive rates of all reactions are spatially uniform throughout that small volume element. If this is true, then the intensive production rate of each species must be spatially uniform throughout that small volume element. If these things are true, then for the small volume element, V_k , we can calculate an extensive production rate for species j in that volume element, $G_{j,k}$ as follows:

$$G_{j,k} = R_{j,k} V_k [=] \frac{\text{moles of } j}{\text{time}}$$

Since we know the extensive production rate in a single volume element, V_k , we can sum up the extensive rate in each element to calculate the total extensive production rate of j throughout the entire volume, V .

$$G_j = \sum_{k=1}^m G_{j,k} = \sum_{k=1}^m R_{j,k} V_k [=] \frac{\text{moles of } j}{\text{time}}$$

Focusing on this part of the equation:

$$G_j = \sum_{k=1}^m R_{j,k} V_k$$

If we assume that we have an infinite number of volume elements ($m \rightarrow \infty$), the volume of each element approaches zero, and this summation becomes an integral:

$$G_j = \int^V R_j dV$$

In other words, we allow that R_j is a **function of Volume**, and we integrate it over the entire reactor volume to compute the extensive production rate of species j , G_j . Without making specific

declarations about the type of reactor we are working with and assumptions about how it operates, we cannot simplify this expression any further. So, for our general species balance on a reactor:

$$\frac{dN_j}{dt} = F_{j,f} - F_j + G_j$$

The final result that we'll use (*for a homogeneous reaction*) is:

$$\frac{dN_j}{dt} = F_{j,f} - F_j + \int^V R_j dV$$

In the next lecture, we will look at how to apply this general balance equation to various types of reactors (Batch, CSTR, PFR) operating under specific assumptions.